

COMSATS University Islamabad, Vehari Campus

Information Communication Technology (CSC101)

Lab 03

Topic: MS EXCEL

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Objective

This lab enables students to understand and use MS Excel. Students will learn how to create spreadsheets, use rows and columns, perform calculations, analyze data, and create reports.

Learning Outcomes

After completing this lab, students will be able to:

- Create spreadsheets
 - Prepare reports
 - Apply data filtering
 - Use formulas and functions
 - Format data conditionally
 - Create charts
 - Perform grouped calculations using Subtotal
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Basic Concept of MS Excel

MS Excel is a spreadsheet program used to store, organize, and analyze numerical data. It is widely used in business, finance, education, and research.

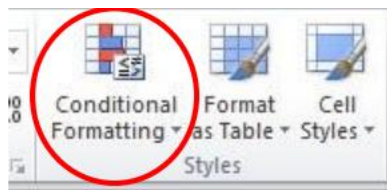
Main Uses of Excel

- Data entry and storage
- Data management
- Accounting
- Financial analysis

- Charting and graphing
- Task and time management
- Financial modeling
- Customer data organization
- Report preparation

ACTIVITY 1 — Conditional Formatting

Conditional formatting automatically changes cell appearance based on a condition.

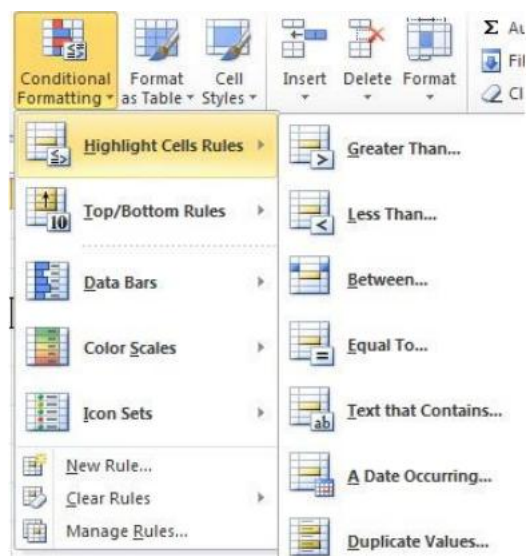


➤ Highlight Cells Rules

1. Greater Than

Purpose: Highlight values greater than a specified number. Steps:

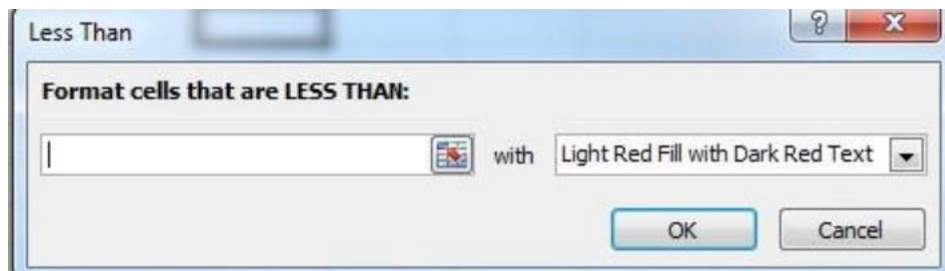
1. Select the data range
2. Go to Home → Conditional Formatting
3. Select Highlight Cells Rules → Greater Than
4. Enter value
5. Select formatting
6. Click OK





2. Less Than

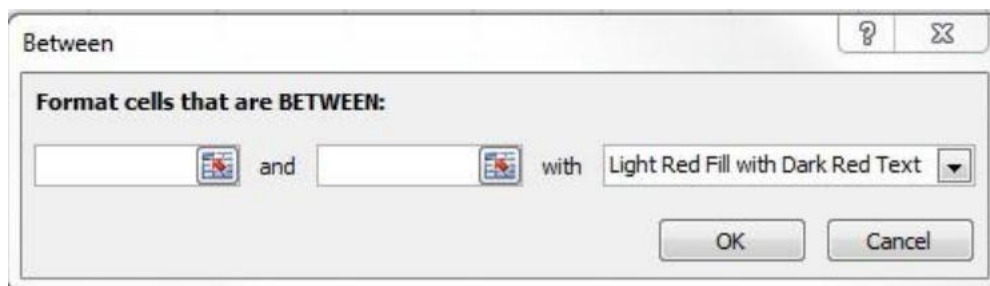
Purpose: Highlight values less than a specified number. Follow the same steps but choose Less Than.



3. Between

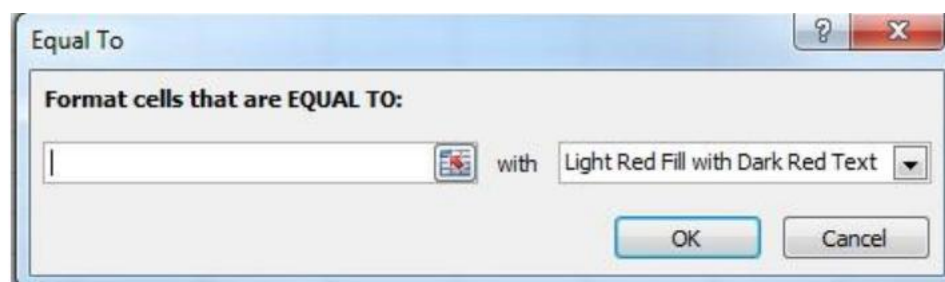
Purpose: Highlight values between two numbers. Steps:

1. Select data range
2. Conditional Formatting → Highlight Cells Rules → Between
3. Enter lower and upper values
4. Select formatting → OK



4. Equal To

Purpose: Highlight cells equal to a specific value.



5. Text That Contains

Purpose: Highlight cells containing specific text.



6. A Date Occurring

Purpose: Highlight cells with specific dates or date ranges.



7. Duplicate Values

Purpose: Highlight duplicate or unique values. Steps:

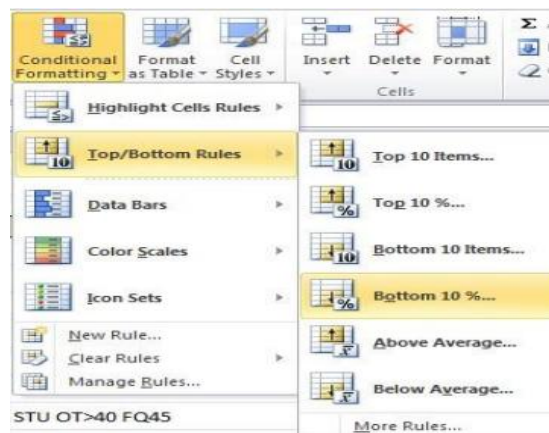
1. Select data
2. Conditional Formatting → Highlight Cells Rules → Duplicate Values
3. Choose Duplicate or Unique → OK



➤ Top / Bottom Rules

Used to identify:

- Top 10 values
- Bottom 10 values
- Top percentage
- Above average
- Below average



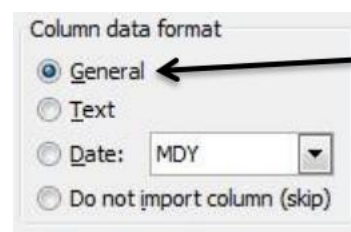
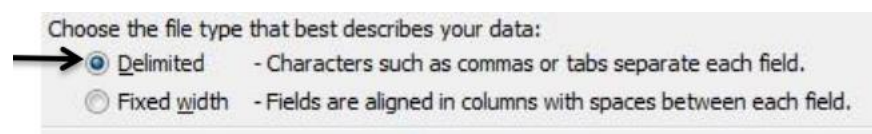
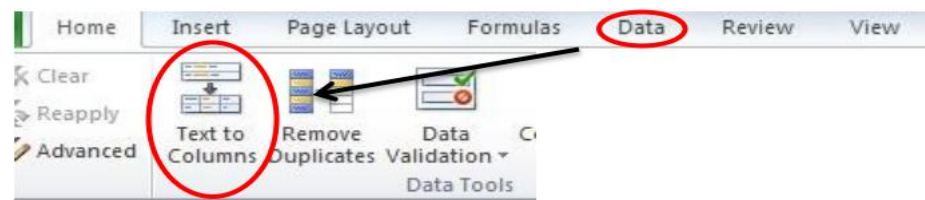
ACTIVITY 2 — Separating Text within a Cell

Used to split combined data into multiple columns.

Steps:

1. Insert a blank column next to the data
2. Select the column containing combined text
3. Go to Data tab → Text to Columns
4. Select Delimited → Next
5. Select delimiter (Space, Comma, etc.)
6. Select data format → Finish

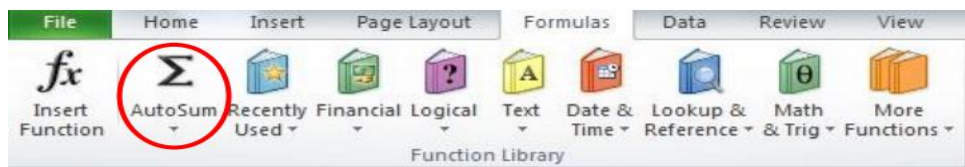
Result: Data will be separated into different columns.



Sallie Shaffer	Sallie	Shaffer
Joseph Garcia	Joseph	Garcia
Sallie Shaffer	Sallie	Shaffer
Surhid Gautam	Surhid	Gautam
Charles Dennis	Charles	Dennis
Yasmine Johnson	Yasmine	Johnson
Kathleen Chilton	Kathleen	Chilton
Kathleen Chilton	Kathleen	Chilton
Helen Martin	Helen	Martin
Helen Sue Martin	Helen	Martin
Joseph Garcia	Joseph	Garcia
Sallie Shaffer	Sallie	Shaffer
Mark Abraham	Mark	Abraham
Mark Abraham	Mark	Abraham
Surhid Gautam	Surhid	Gautam
Carmen Roberts	Carmen	Roberts

ACTIVITY 3 — Functions and Formulas

Functions perform automatic calculations.

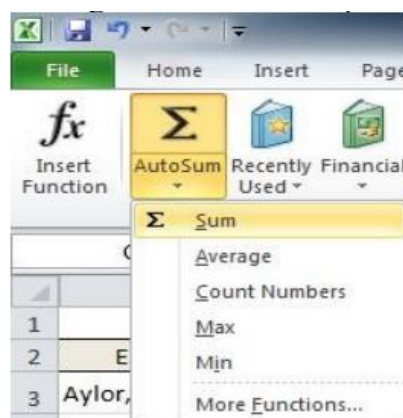


Last Name	First Name	Banner ID	Amount Paid	Date Paid
Smith	John	745082	3,000	9/8/2013
			2,500	10/12/2013
			1,500	12/10/2013
			2,000	3/15/2014
			1,000	6/10/2014

SUM Function

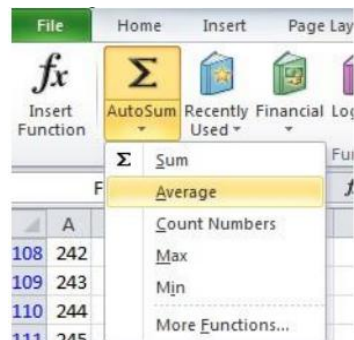
Adds numbers in a range. **Steps:**

1. Select cell below numbers
2. Click AutoSum
3. Press Enter



Other AutoSum Functions

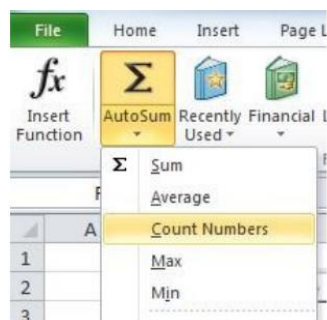
- Average → Finds mean value



Last Name	First Name	Banner ID	Amount Paid	Date Paid
Smith	John	745082	\$3,000	9/8/2013
			\$2,500	10/12/2013
			\$1,500	12/10/2013
			\$2,000	3/15/2015
			\$1,000	6/10/2015
			\$2,000	

- Count Numbers → Counts numeric entries

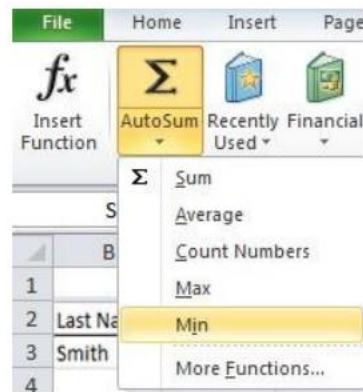
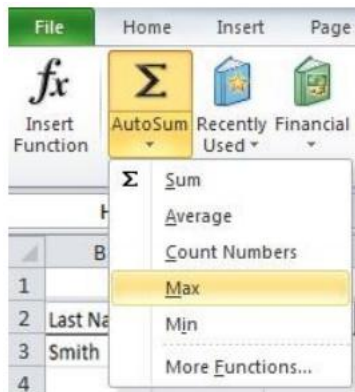
Last Name	First Name	Banner ID	Amount Paid	Date Paid
Smith	John	745082	3000	9/8/2013
			2500	10/12/2013
			1500	12/10/2013
			2000	3/15/2015
			1000	6/10/2015



Last Name	First Name	Banner ID	Amount Paid	Date Paid
Smith	John	745082	3000	9/8/2013
			2500	10/12/2013
			1500	12/10/2013
			2000	3/15/2015
			1000	6/10/2015
			5	

- Max → Finds highest value
- Min → Finds lowest value

Last Name	First Name	Banner ID	Amount Paid	Date Paid
Smith	John	745082	3000	9/8/2013
			2500	10/12/2013
			1500	12/10/2013
			2000	3/15/2015
			1000	6/10/2015



Subtotal Feature

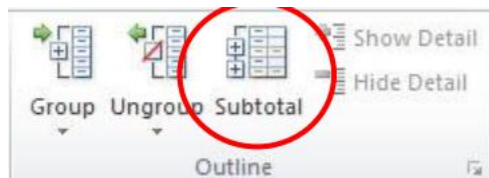
Used to calculate totals for grouped data automatically.

Last Name	First Name	Banner ID	Amount Paid	Semester
Smith	John	745082	\$ 3,000.00	Fall
			\$ 2,500.00	Fall
			\$ 1,500.00	Fall
			\$ 2,000.00	Spring
			\$ 1,000.00	Spring

➤ One Level Subtotal

Steps:

1. Select dataset
2. Go to Data → Subtotal
3. At Each Change In → select category
4. Use Function → Sum
5. Add Subtotal To → select column
6. Click OK



Subtotal

At each change in:
Semester

Use function:
Sum

Add subtotal to:

- ☐ Last Name
- ☐ First Name
- ☐ Banner ID
- ☒ Amount Paid
- ☐ Semester

☒ Replace current subtotals

☐ Page break between groups

☒ Summary below data

Remove All OK Cancel

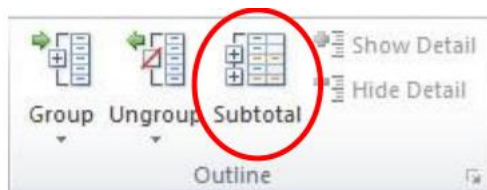
	A	B	C	D	E	F
1	Last Name	First Name	Banner ID	Amount Paid	Semester	
2	Smith	John	745082	\$ 3,000.00	Fall	
3				\$ 2,500.00	Fall	
4				\$ 1,500.00	Fall	
5				\$ 7,000.00	Fall Total	
6				\$ 2,000.00	Spring	
7				\$ 1,000.00	Spring	
8				\$ 3,000.00	Spring Total	
9				\$ 10,000.00	Grand Total	

➤ Nested Subtotal

Used for multi-level grouping. **Steps:**

1. Apply first subtotal
2. Open Subtotal again
3. Select second grouping field
4. Uncheck Replace Current Subtotals
5. Click OK

Semester	Last Name	First Name	Banner ID	Amount Paid
Fall 2013	Smith	John	745082	\$ 3,000.00
Fall 2013	Smith	John	745082	\$ 2,500.00
Fall 2013	Smith	John	745082	\$ 1,500.00
Fall 2013	Jones	Katherine	642986	\$ 1,500.00
Fall 2013	Jones	Katherine	642986	\$ 2,000.00
Fall 2013	Jones	Katherine	642986	\$ 3,250.00
Spring 2014	Smith	John	745082	\$ 2,000.00
Spring 2014	Smith	John	745082	\$ 1,000.00
Spring 2014	Jones	Katherine	642986	\$ 3,000.00
Spring 2014	Jones	Katherine	642986	\$ 2,750.00
Fall 2014	Smith	John	745082	\$ 2,750.00
Fall 2014	Smith	John	745082	\$ 3,250.00
Fall 2014	Jones	Katherine	642986	\$ 1,750.00
Fall 2014	Jones	Katherine	642986	\$ 2,000.00
Spring 2015	Smith	John	745082	\$ 1,750.00
Spring 2015	Smith	John	745082	\$ 2,250.00
Spring 2015	Smith	John	745082	\$ 2,000.00
Spring 2015	Jones	Katherine	642986	\$ 2,250.00
Spring 2015	Jones	Katherine	642986	\$ 2,500.00
Spring 2015	Jones	Katherine	642986	\$ 3,000.00



Subtotal

At each change in:
Semester

Use function:
Sum

Add subtotal to:

- ☐ Semester
- ☐ Last Name
- ☐ First Name
- ☐ Banner ID
- ☒ Amount Paid

☐ Replace current subtotals

☐ Page break between groups

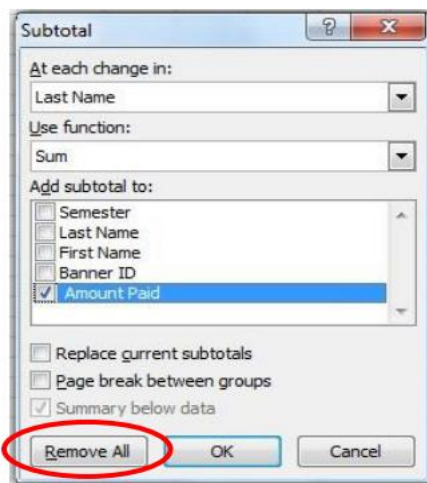
☒ Summary below data

Remove All OK Cancel

1	2	3		A	B	C	D	E
				Semester	Last Name	First Name	Banner ID	Amount Paid
				Fall 2013	Smith	John	745082	\$ 3,000.00
				Fall 2013	Smith	John	745082	\$ 2,500.00
				Fall 2013	Smith	John	745082	\$ 1,500.00
				Fall 2013	Jones	Katherine	642986	\$ 1,500.00
				Fall 2013	Jones	Katherine	642986	\$ 2,000.00
				Fall 2013	Jones	Katherine	642986	\$ 3,250.00
				Fall 2013 Total				\$ 13,750.00
				Spring 2014	Smith	John	745082	\$ 2,000.00
				Spring 2014	Smith	John	745082	\$ 1,000.00
				Spring 2014	Jones	Katherine	642986	\$ 3,000.00
				Spring 2014	Jones	Katherine	642986	\$ 2,750.00
				Spring 2014 Total				\$ 8,750.00
				Fall 2014	Smith	John	745082	\$ 2,750.00
				Fall 2014	Smith	John	745082	\$ 3,250.00
				Fall 2014	Jones	Katherine	642986	\$ 1,750.00
				Fall 2014	Jones	Katherine	642986	\$ 2,000.00
				Fall 2014 Total				\$ 9,750.00
				Spring 2015	Smith	John	745082	\$ 1,750.00
				Spring 2015	Smith	John	745082	\$ 2,250.00
				Spring 2015	Smith	John	745082	\$ 2,000.00
				Spring 2015	Jones	Katherine	642986	\$ 2,250.00
				Spring 2015	Jones	Katherine	642986	\$ 2,500.00
				Spring 2015	Jones	Katherine	642986	\$ 3,000.00
				Spring 2015 Total				\$ 13,750.00
				Grand Total				\$ 46,000.00

➤ Remove Subtotal

Data → Subtotal → Remove All



GRADED LAB TASKS

Complete all tasks in MS Excel.

✓ LAB TASK 1 — Sales Invoice

Create a formatted invoice with at least 4 items.

Columns: Item | Quantity | List Price | Discount | Your Price | Total

Formulas: Your Price = List Price – Discount Total = Quantity × Your Price Subtotal =

SUM of Total column GST = Subtotal × 17% Amount Due = Subtotal + GST

Requirements:

- Apply proper formatting
- Use formulas only
- Include at least four products

✓ LAB TASK 2 — CSC101 Grade Book

Create the following table: Name | Exam1 | Exam2 | Exam3 | Final | Average | Letter Grade

Steps:

- Enter all student data
- Calculate Average = AVERAGE function
- Sort names alphabetically
- Assign Letter Grade using IF function
- Grade Criteria: Score $\geq 90 \rightarrow A$ Score $\geq 80 \rightarrow B$ Score $\geq 70 \rightarrow C$ Score $\geq 60 \rightarrow D$ Score $< 60 \rightarrow F$
- Example-IF-Formula:
`=IF(F2>=90,"A",IF(F2>=80,"B",IF(F2>=70,"C",IF(F2>=60,"D","F"))))`
- Create a Bar Chart of Final Grades

✓ LAB TASK 3 — Class Make-up Charts**Chart 1: Class Level Distribution**

Freshmen = 10 Sophomore = 8 Junior = 4 Senior = 1 No Response = 5 Create a chart.

Chart 2: Gender Distribution

Men = 11 Women = 17 Create a Pie Chart.

Chart 3: Homework Completion

Students completed homework = 23 Total students = 28 Formula: Percentage = $(23 / 28) \times 100$ Create a Pie Chart.

✓ LAB TASK 4 — Vacation Budget

Step 1: Planning (Word Document)

Write:

- Destination
- Activities
- Estimated budget

Step 2: Excel Budget Sheet

Create categories:

- Transportation
- Accommodation
- Food
- Sightseeing
- Shopping
- Other Activities

Step 3: Calculations

Category totals Grand total Contingency = Total × 10% Final Budget = Total + Contingency

✓ Lab Task Solutions:

LAB TASK 1 — Sales Invoice (Solved Structure + Formulas)

Step 1 — Create Table

A	B	C	D	E	F
Item	Quantity	List Price	Discount	Your Price	Total

Enter at least 4 items.

Step 2 — Formulas

Your Price

Cell E2:

=C2-D2

Copy down.

Total

Cell F2:

=B2*E2

Copy down.

Step 3 — Subtotal

Below Total column:

=SUM(F2:F5)

Step 4 — GST (17%)

If Subtotal is in F6:

=F6*17%

Step 5 — Amount Due

=F6+F7

Step 6 — Formatting

- ✓ Bold headings
- ✓ Currency format
- ✓ Borders
- ✓ Alignment

✓ LAB TASK 2 — CSC101 Grade Book (Complete Solution)

Step 1 — Table Structure

A	B	C	D	E	F	G
Name	Exam1	Exam2	Exam3	Final	Average	Letter Grade

Enter given data.

Step 2 — Average Formula

Cell F2: =AVERAGE(B2:E2)

Copy down.

Step 3 — Letter Grade Formula

Cell G2: =IF(F2>=90,"A",IF(F2>=80,"B",IF(F2>=70,"C",IF(F2>=60,"D","F"))))

Copy down.

Step 4 — Alphabetical Sorting

1. Select table
2. Data → Sort
3. Sort by Name → A to Z

Step 5 — Bar Chart

1. Select Name + Final columns
 2. Insert → Bar Chart
 3. Add chart title
-

✓ LAB TASK 3 — Charts (Solved)

Chart 1 — Class Level Distribution

Create table:

Level	Students
Freshmen	10
Sophomore	8
Junior	4
Senior	1
No Response	5

Steps:

1. Select table
2. Insert → Column or Pie Chart

Chart 2 — Gender Pie Chart

Gender	Students
Men	11
Women	17

Steps:

1. Select data
2. Insert → Pie Chart

Chart 3 — Homework Completion

Percentage Formula = $23/28*100$

Create table:

Status	Students
Completed	23
Not Completed	5

Insert Pie Chart.

✓ LAB TASK 4 — Vacation Budget (Solved Model)

Step 1 — Example Plan

Destination: Any city

Activities: sightseeing, food, shopping

Step 2 — Budget Table

Category	Cost
Transport	500
Accommodation	800
Food	300
Sightseeing	200
Shopping	250

Step 3 — Total Cost

=SUM(B2:B6)

Step 4 — Contingency (10%)

If total in B7:

=B7*10%

Step 5 — Final Budget

=B7+B8

End of Lecture Lab 03 (MS EXCEL)